Bilkent University EE313 ECD
Lab 2 Report
Low-Dropout Voltage Regulator

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Introduction and Aim

In this lab, our aim is to firstly find a method to measure β value of a pnp transistor. Then, we were required to design a low-dropout(LDO)(0.7V max) voltage regulator with an output current of 100mA. If the output is the value we wanted, a greed LED must turn ON. Out output voltage values are determined by our Bilkent ID numbers. V_{out}= 7.5+mod(BilkentID,6). After all that, we will start building these circuits on hardware using the calculated component values on the software part. The 3 specifications must also be true for the hardware part. These results will be compared.

Software Part:

Part A:

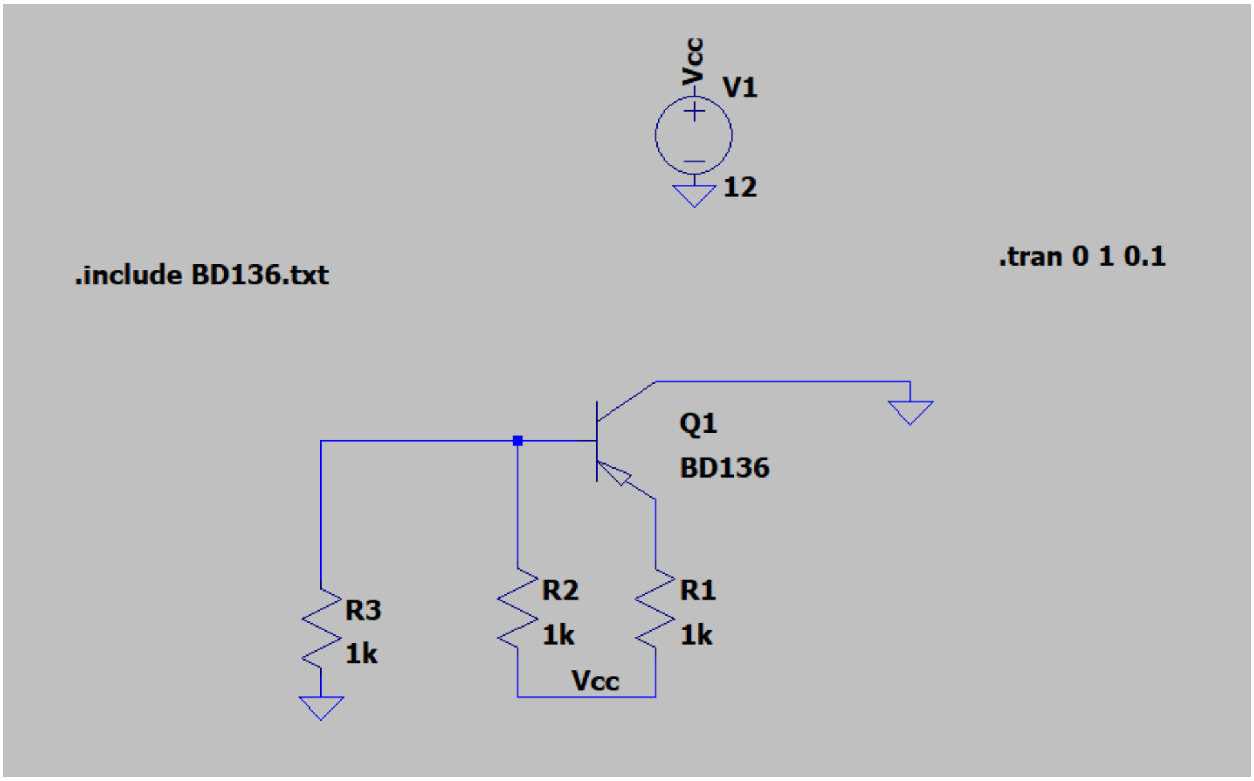
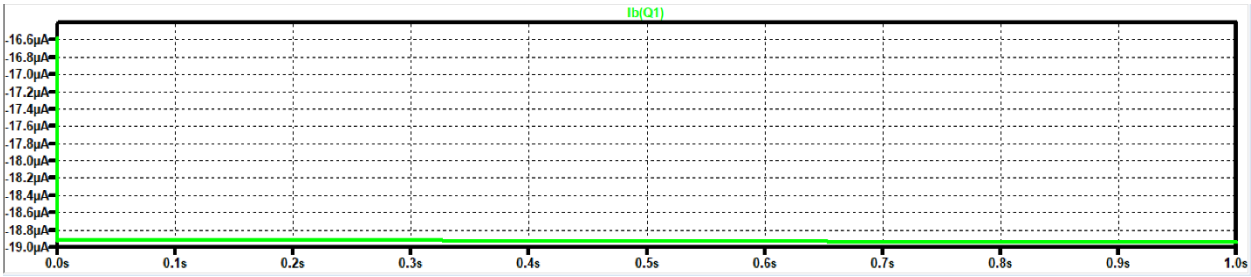


Figure 1: Circuit diagram to measure β of a pnp diode.

The formula to calculate the β value is given below.

$$\beta = I_C / I_B$$

So we need to measure I_C *and* I_B.



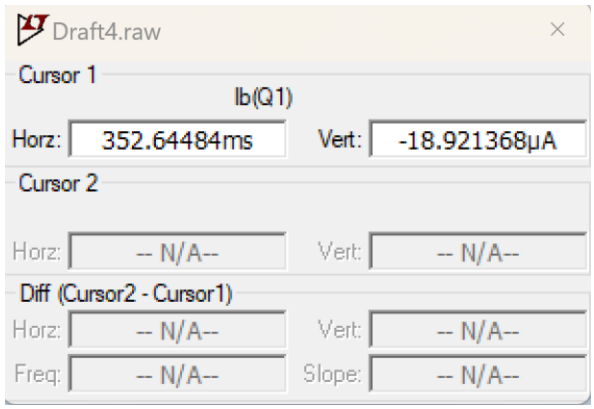
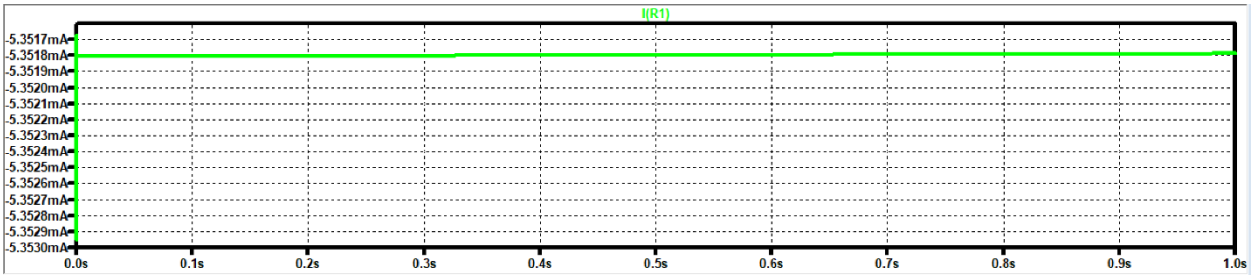


Figure 2: I_B measurement.



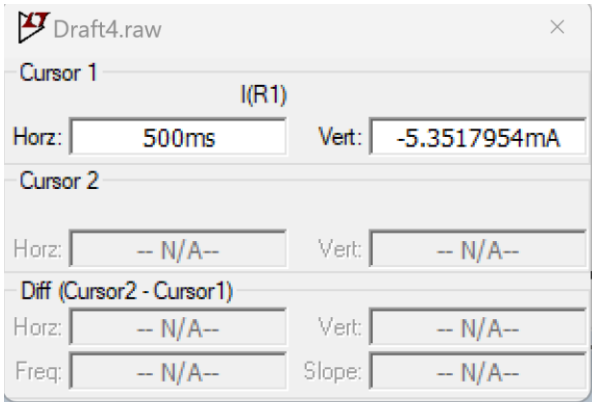


Figure 3: I_S measurement.

The formula is used to calculate β:

$$\beta = \frac{5.35}{0.01892} \approx 282$$

Part B:

Specifications

1. Line regulation:
   When V_{in} is varying between V_{out}+0.7 to V_{out}+6 at 100Hz, the output voltage, Vout, changes by no more than 20mV when the output current is 100mA (R_L=V_{out}/0.1).
2. Load regulation:
   When V_{in}=V_{out}+2, the output voltage, V_{out}, changes no more than 20mV when the output cur rent changes between 0mA and 100mA at 100Hz. (In LTSpice, you can connect a sinusoidal current source at the output varying between 0 and 100mA.)
3. A green LED should turn ON if the regulation is achieved. Otherwise, it should turn off, for example, because the input voltage is too low or the output current is too high.

Here is the designed circuit with calculated component values. BD136 is used as a power BJT and for the OPAMP, LM358 is used as requested. I chose 1N750 as the Zener diode.

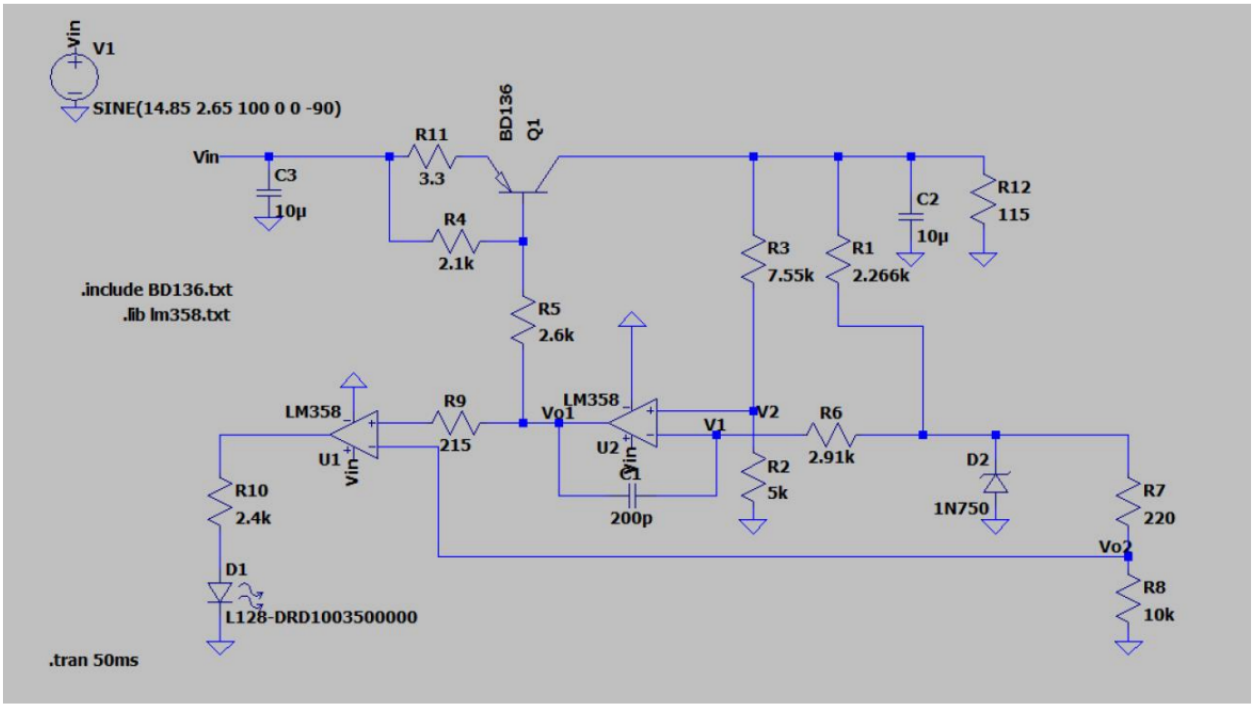


Figure 4 Schematic of the design on LTSpice

Firstly, V_{out} is chosen as 11.5 since V_{out} = 7.5+mod([ID] ,6). A Zener diode that has a breakdown voltage value of 4.7V is also used in this circuit and the values of the components are calculated according to the instructions we were given. The value $C_1 = 200pF$ is chosen to reduce the oscillation on the output voltage since the input voltage changes rapidly. Also, here are the waveforms of the currents on R_2, R_3 and R_7, R_8. The currents on them must be less than 1mA and 0.5mA respectively. The values of them are chosen according to this specification.

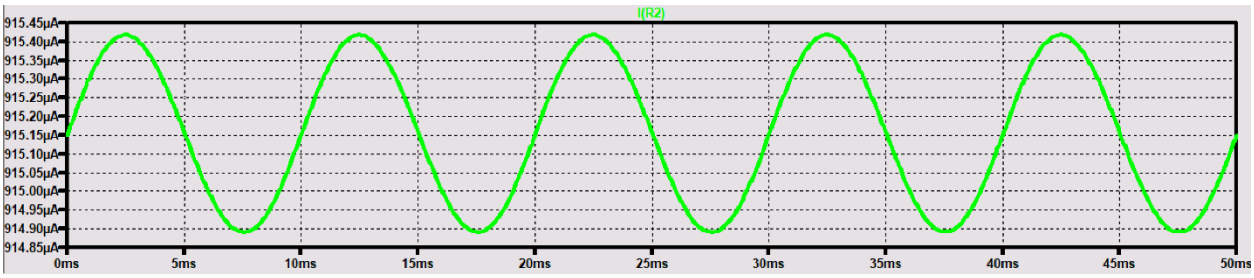


Figure 5: Current on R2

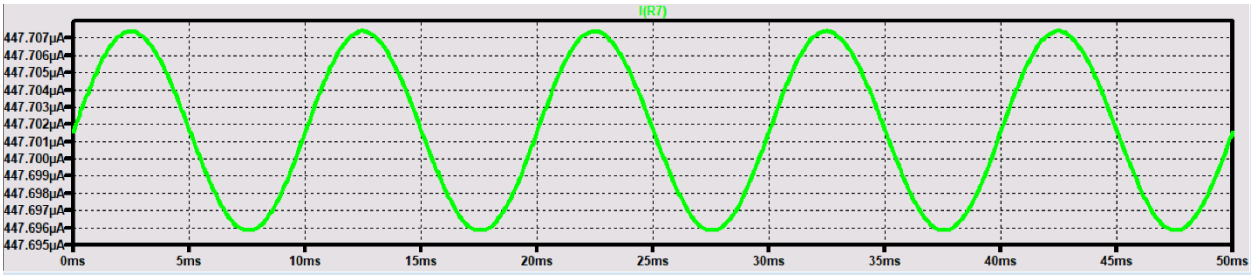


Figure 6: Current on R7

Satisfaction of 3 Criteria and Results

1. **V_{in} must be between 12.2V and 17.5V at 100Hz. A phase of -90 degrees is added for voltage to start from the lower value. Also R_L is calculated as 115Ω. The output voltage should not change more than 20mV when the output current is 100mA.**

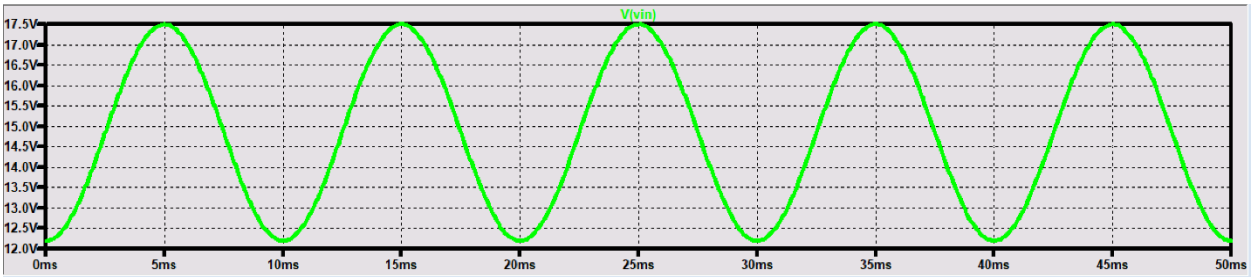


Figure 7 Vin

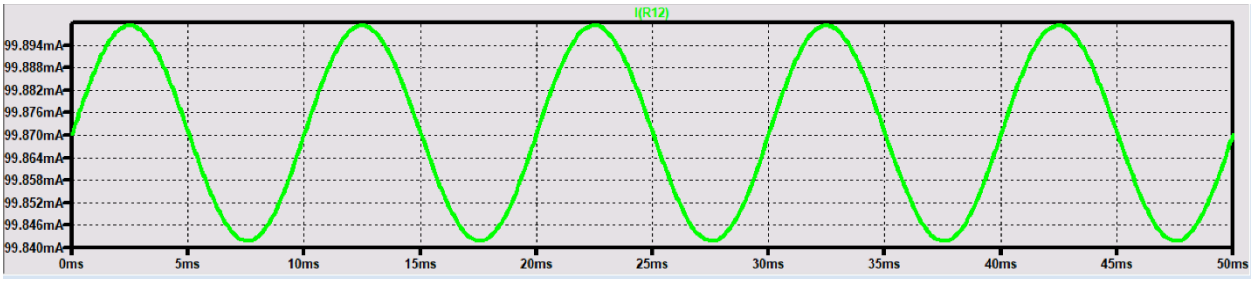


Figure 8: Output Current

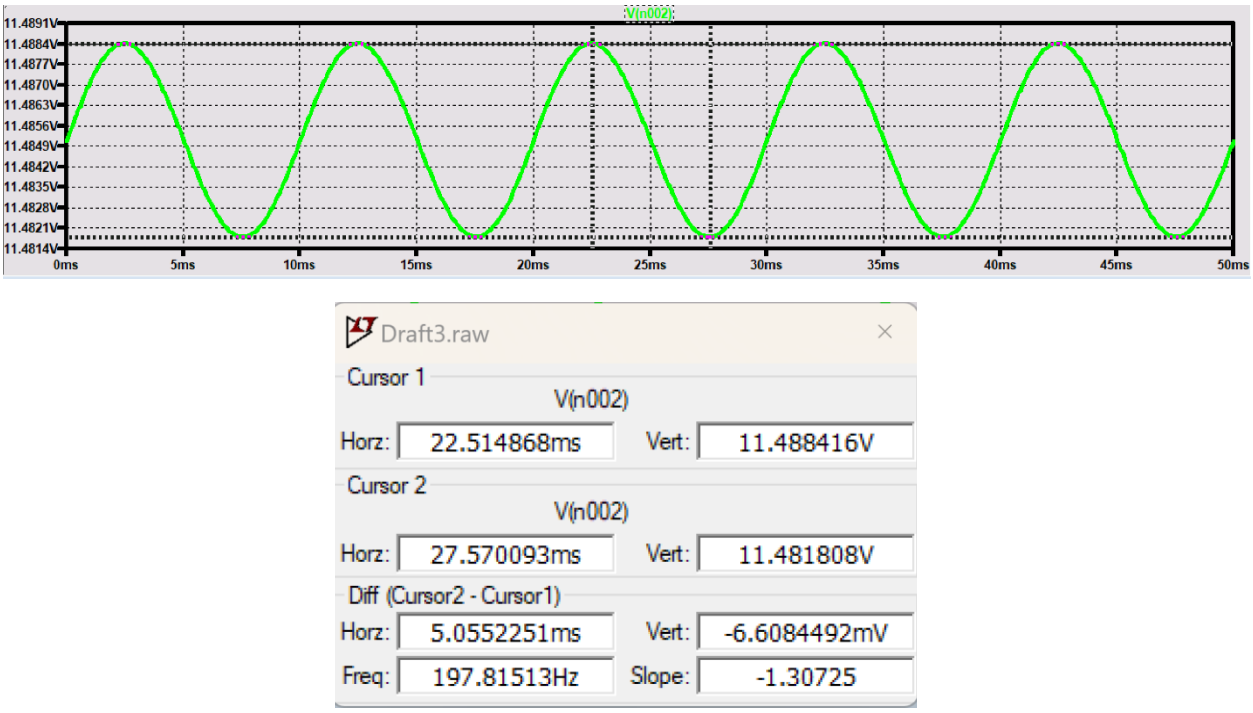


Figure 9: Output Voltage

Since it can be seen from the waveforms, the output current V_{out} is between 11.488V and 11.481V and the difference is at max 6.60mV when the output current is 100mA. Since it is less than 20mV this requirement is satisfied.

2. **When $V_{in} = V_{out} + 2$, the output voltage again changes no more than 20mV when the output current is changing between 0 and 100mA at 100Hz.**

   To make this, a current source is connected in the place of the load resistor and V_{in} is now 13.5V. Again, a phase of -90 degree is added to the current source to see the current start from 0mA.

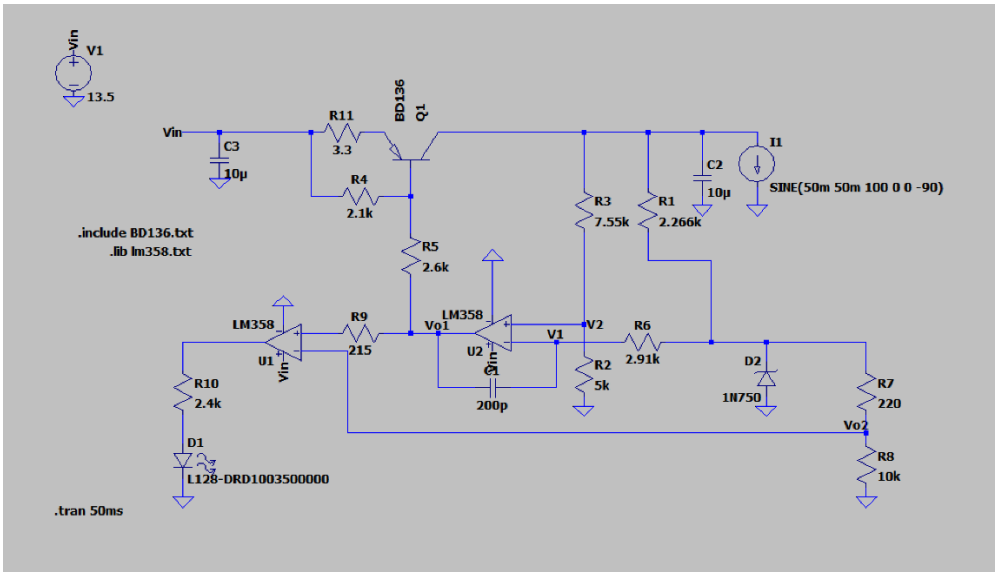


Figure 10: The new schematic for measurement purposes

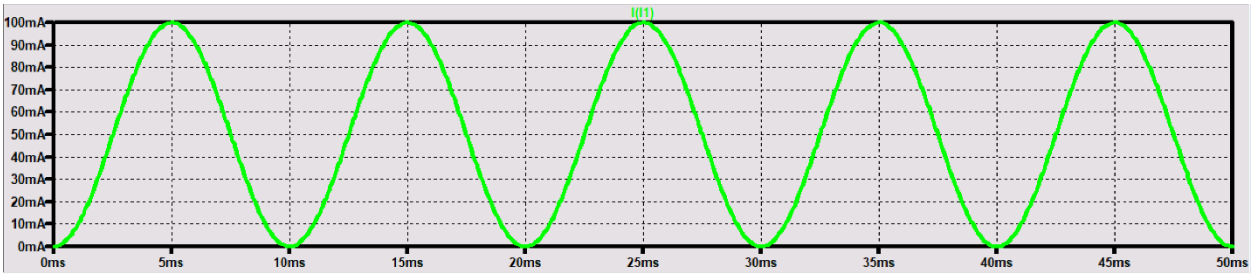


Figure 11: Output current

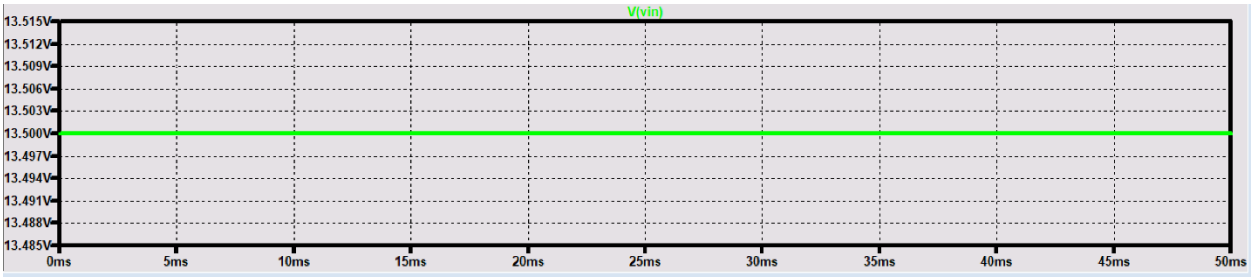


Figure 12: Input Voltage

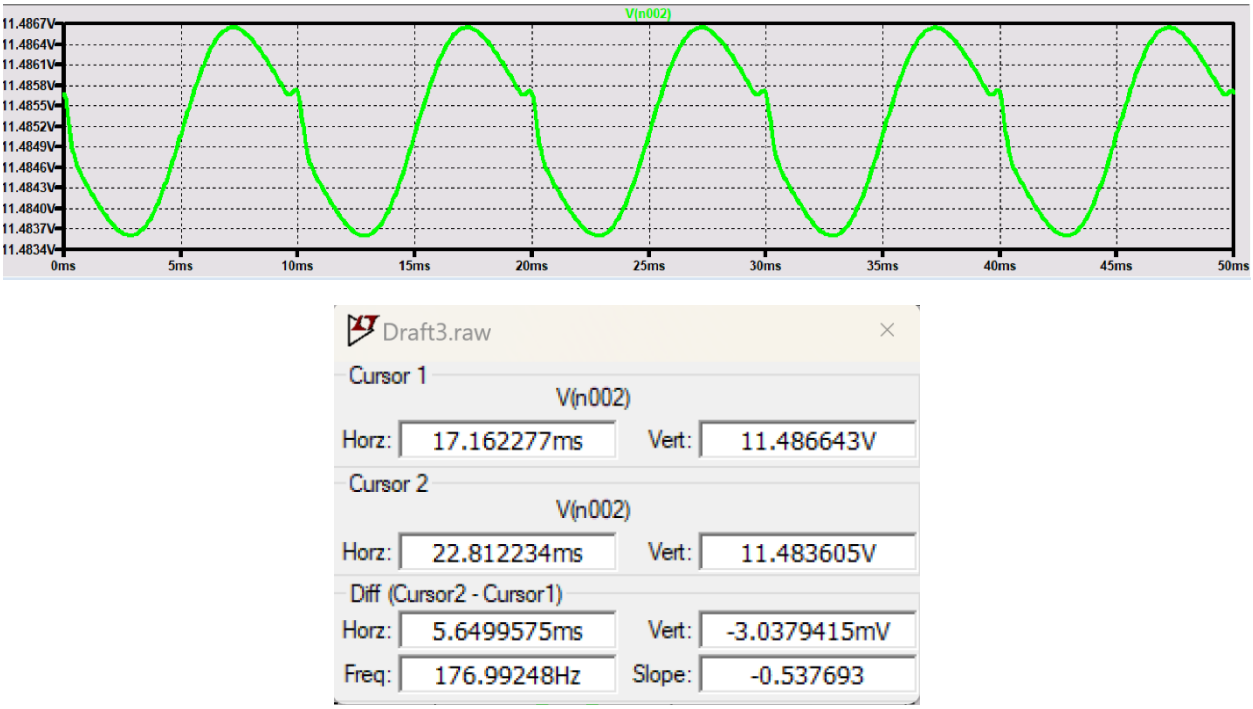


Figure 13: Output Voltage

As it can be seen from the waveforms, when the output current is set between 0mA and 100mA and the input voltage is 13.5V, the output voltage is still in the specification range. It is varying between 11.4866V and 11.4836V with a max change of 3.0379mV which is less than 20mV. Therefore, this requirement is also satisfied.

3. **If regulation is achieved, a greed LED must turn ON. If not, it must turn OFF. To turn OFF, the input voltage must be too low, or the output current is too high.**

To test this, I tried both cases where the input voltage is low and the output current is too high to reach regulation. Firstly, to test it with low voltages, I changes the input voltage source to create a voltage from 0V to 17.5V. This way, we can see where the regulation has started. To see the regulation, we can check the current on the LED. It will be really close to 0mA where the regulation is not happening.

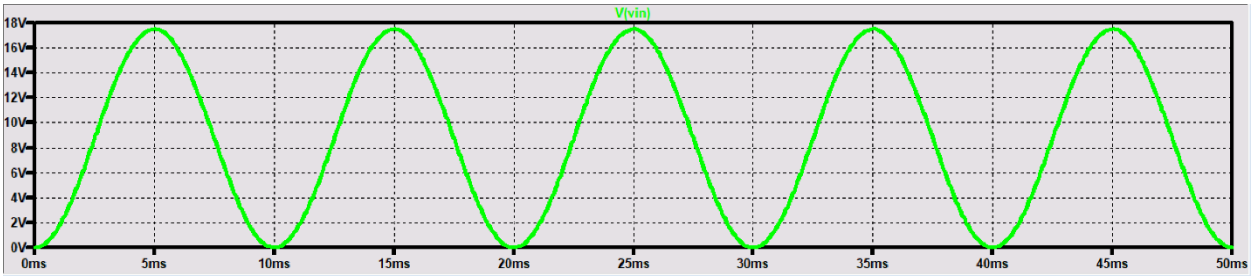


Figure 14: Vin to test regulation between 0V-17.5V

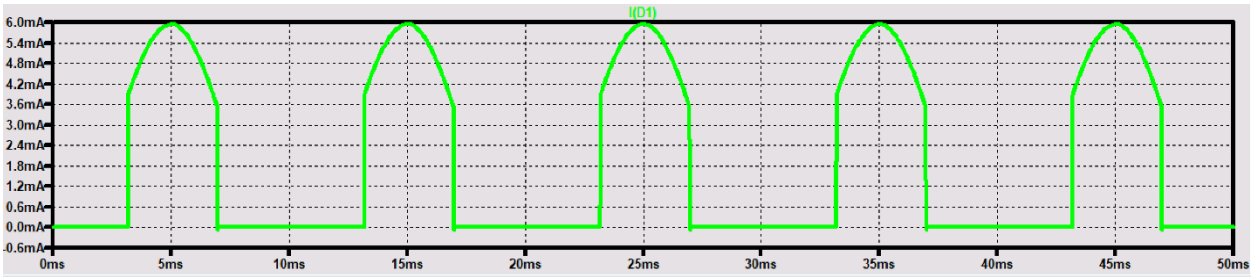


Figure 15: Current on the LED

By these waveforms, we can clearly say that when the input voltage is lower than expected, the LED turns OFF and when the input voltage is in expected range, the LED turns ON which also states that the regulation is happened correctly.

Here is the test with higher output currents. To test this, I changed the output current source to create a current between 0mA and 250mA. This way we can see where the regulation started to happen. We will also look at the LED current to see the regulation. Input voltage is chosen as 13.5 in this situation.

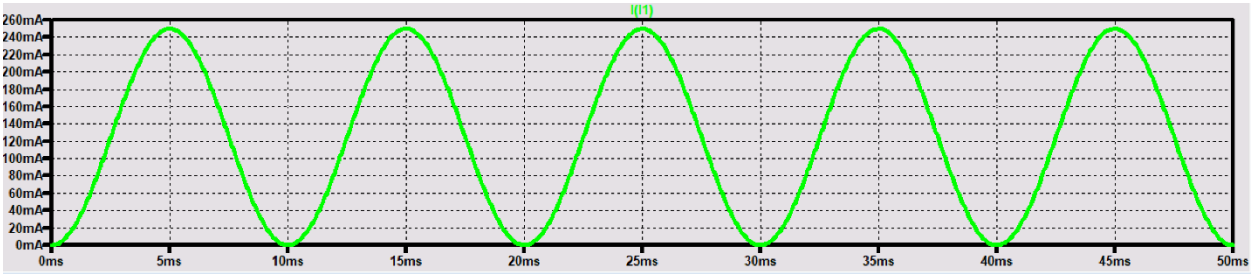


Figure 16: Output current between 0mA-250mA to check regulation.

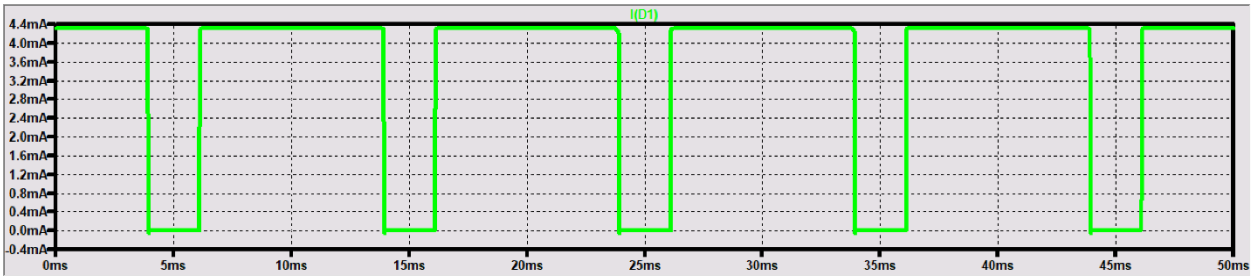


Figure 17: Current on the LED

Again, we can see that on higher values of current, the LED goes OFF which says that the regulation is not achieved. The LED goes ON when it is not high so this requirement is also satisfied.

We were also required to calculate the T_J and T_C values according to the formulas given.

$$T_J = T_A + R_{\theta JA}P_D \; and \; T_C = T_J - R_{\theta JC}P_D$$

Where $T_A = 25\,℃\ and\ P_D = 0.3W$. The values of $R_{\theta JA}$ and $R_{\theta JC}$ can be found on the datasheet.

THERMAL CHARACTERISTICS

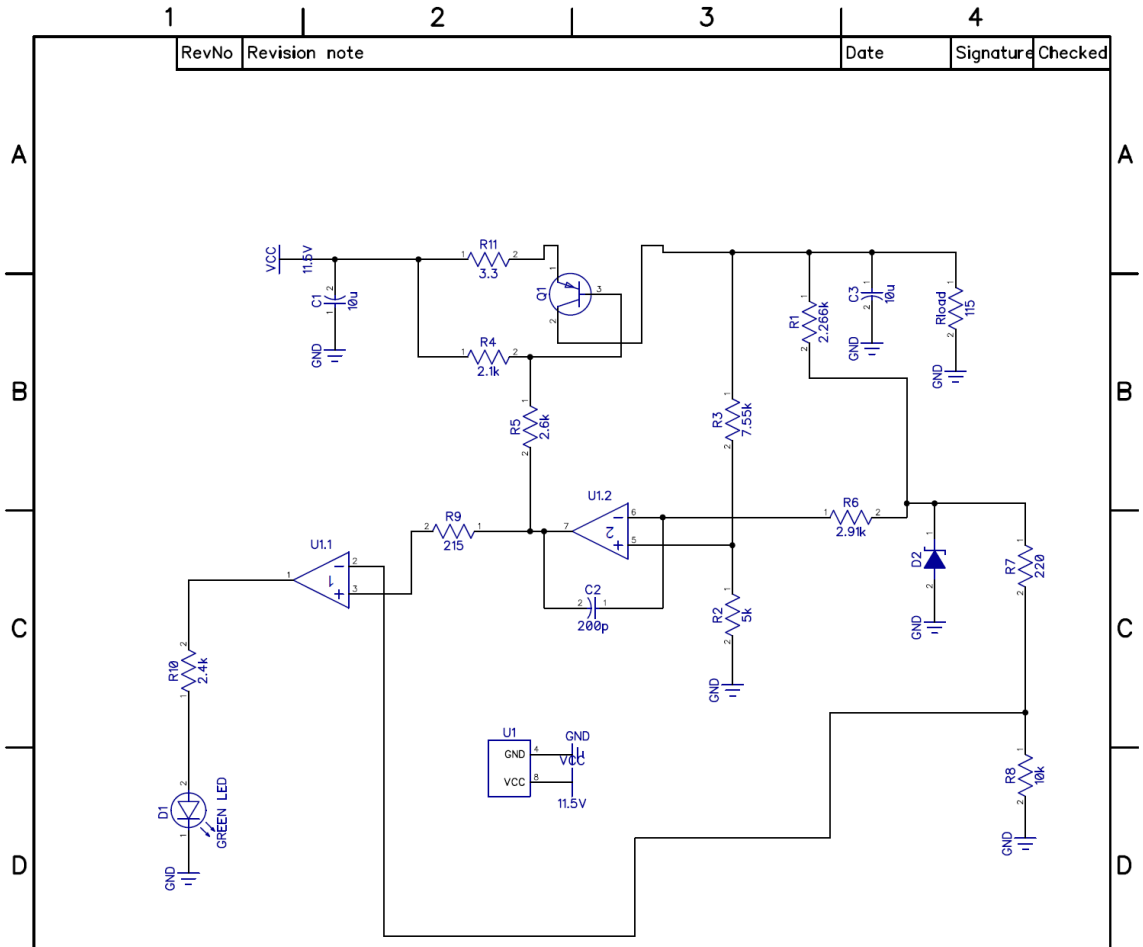
Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction−to−Case	$R_{\theta JC}$	10	°C/W
Thermal Resistance, Junction−to−Ambient	$R_{\theta JA}$	100	°C/W

Figure 18: Thermal Characteristics Values.

After using the formulas given above, we can calculate:

$$T_J = 55℃\ and\ T_C = 52℃$$

Schematic:



#	RefDes	Value	Name
1	C1, C3	10u	CAP100
2	C2	200p	CAP100
3	D1		GREEN LED
4	D2		1N750A-1
5	Q1		BD136G
6	R1	2.266k	CFR25SJR-52-100K
7	R2	5k	CFR25SJR-52-100K
8	R3	7.55k	CFR25SJR-52-100K
9	R4	2.1k	CFR25SJR-52-100K
10	R5	2.6k	CFR25SJR-52-100K
11	R6	2.91k	CFR25SJR-52-100K
12	R7	220	CFR25SJR-52-100K
13	R8	10k	CFR25SJR-52-100K
14	R9	215	CFR25SJR-52-100K
15	R10	2.4k	CFR25SJR-52-100K
16	R11	3.3	CFR25SJR-52-100K
17	Rload	115	CFR25SJR-52-100K
18	U1		LM358N

Title: Low Dropout Voltage Regulator		
Size	Number	Rev
Date: 12.10.2024		Ozgur Ekin Sonmez
Filename		Sheet

Hardware Part:

Now, for the first part, the measurement of β the function of the multimeter is used. When the multimeter is in hFE mode, it can also compute the β value of our transistor. Here, in this case, my value measured was 266. The software result was 282, so there is an error of 5.67%. Here is the measurement:

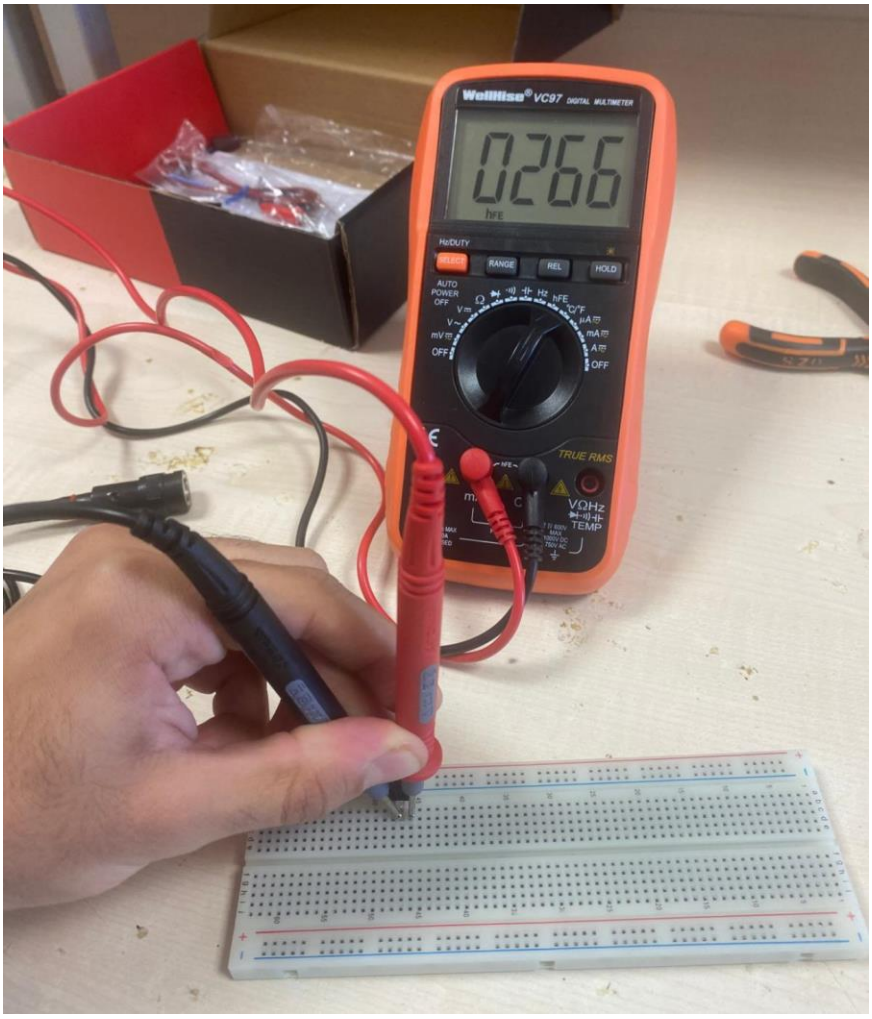


Figure 19: β measurement.

Now, for the second part, the voltage regulator circuit is builded using the component values calculated above. Here is the image of the circuit builded:

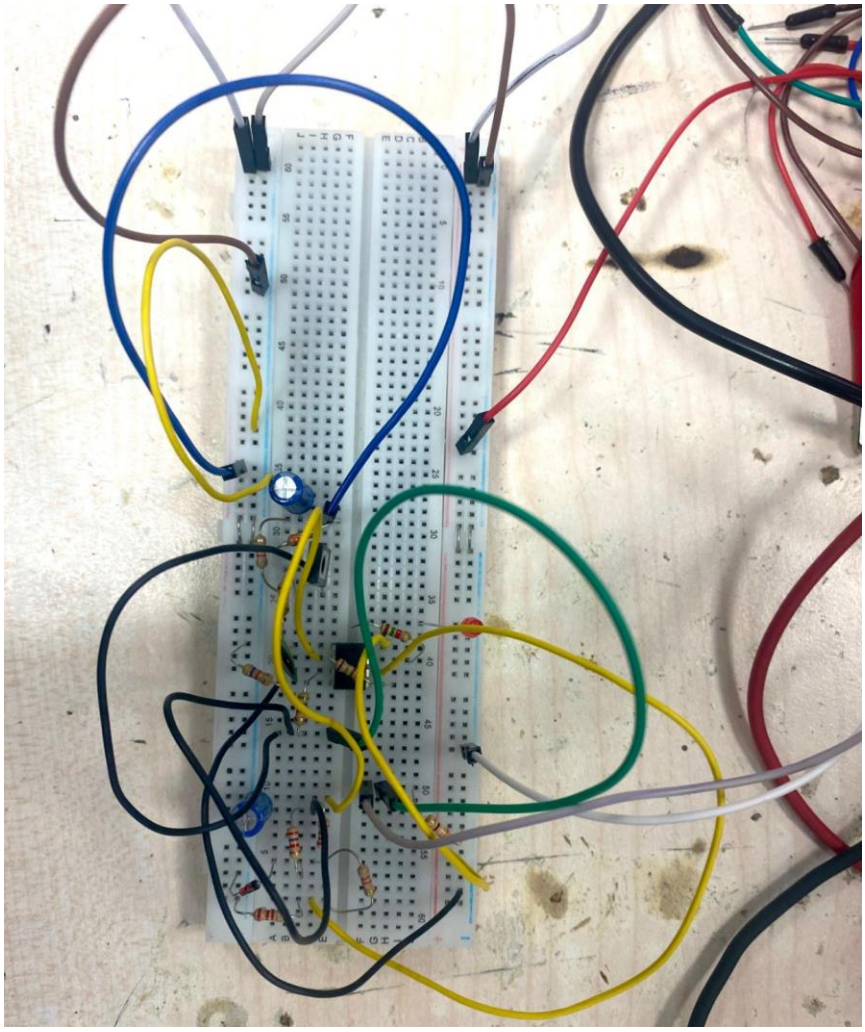


Figure 20: Hardware Regulator Circuit

To test this circuit, I used the voltage source. The input voltage values for my circuit was between 12.2V and 17.5V. So the circuit is tested between these values. The load resistor value is adjusted to see the wanted output voltage at first place. It is increased to 560Ωs.

1) Line regulation. Between our input voltage values, we should not see a change on the output voltage. The change should be at max 20mV. This is tested as shown in the figures below with some different increased input voltages:

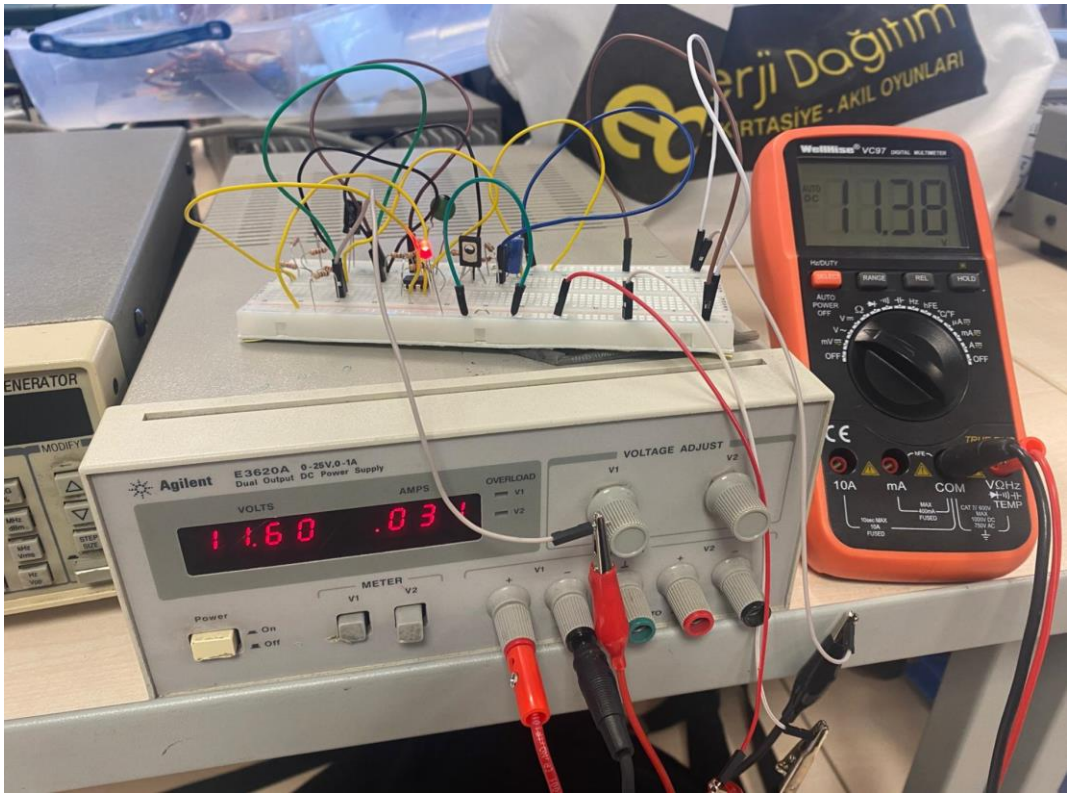


Figure 21: Regulation is achieved under 11.6V

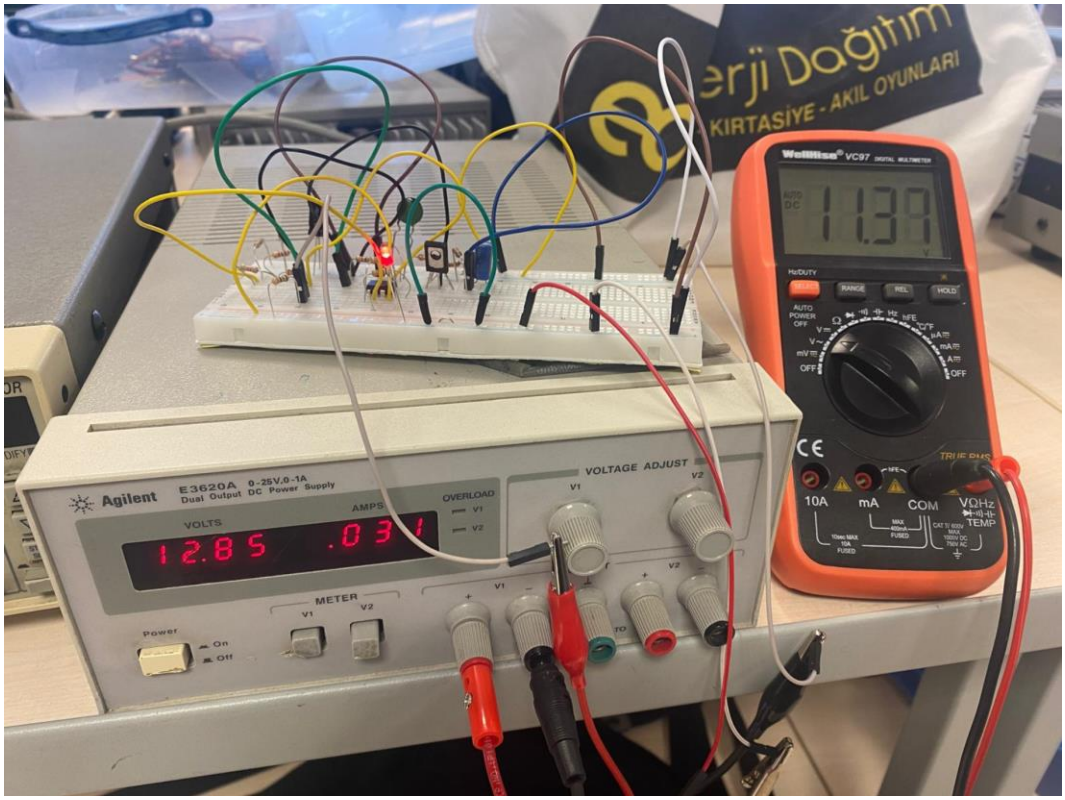


Figure 22: Regulation is achieved under 12.85V

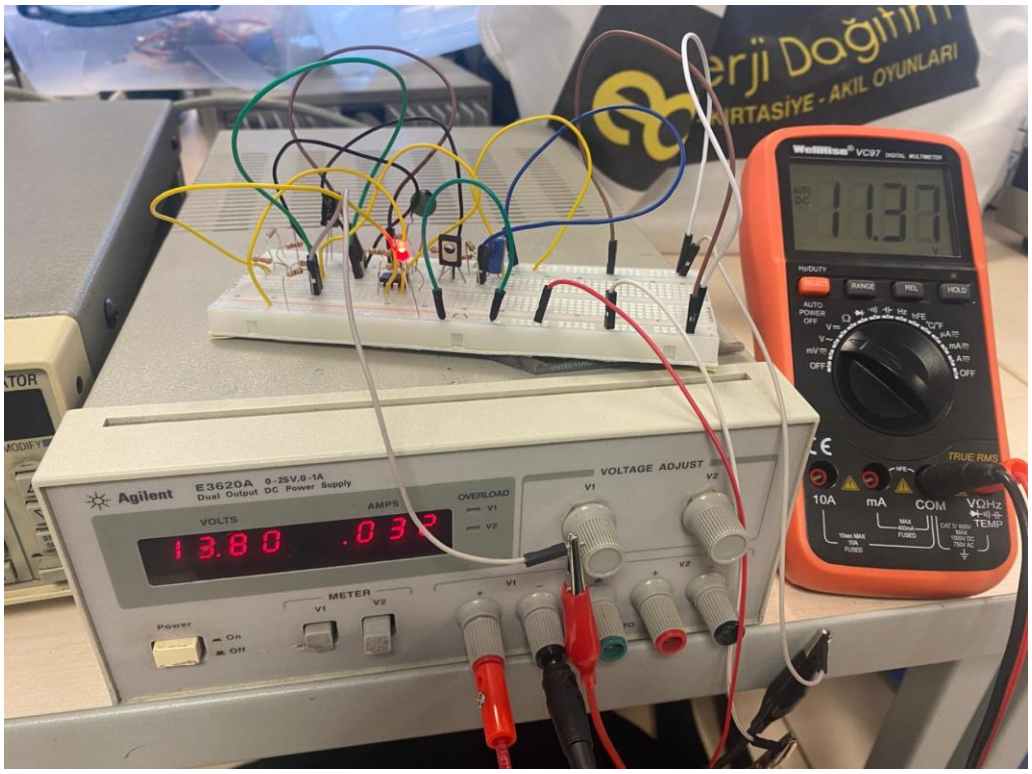


Figure 23: Regulation is achieved under 13.8V

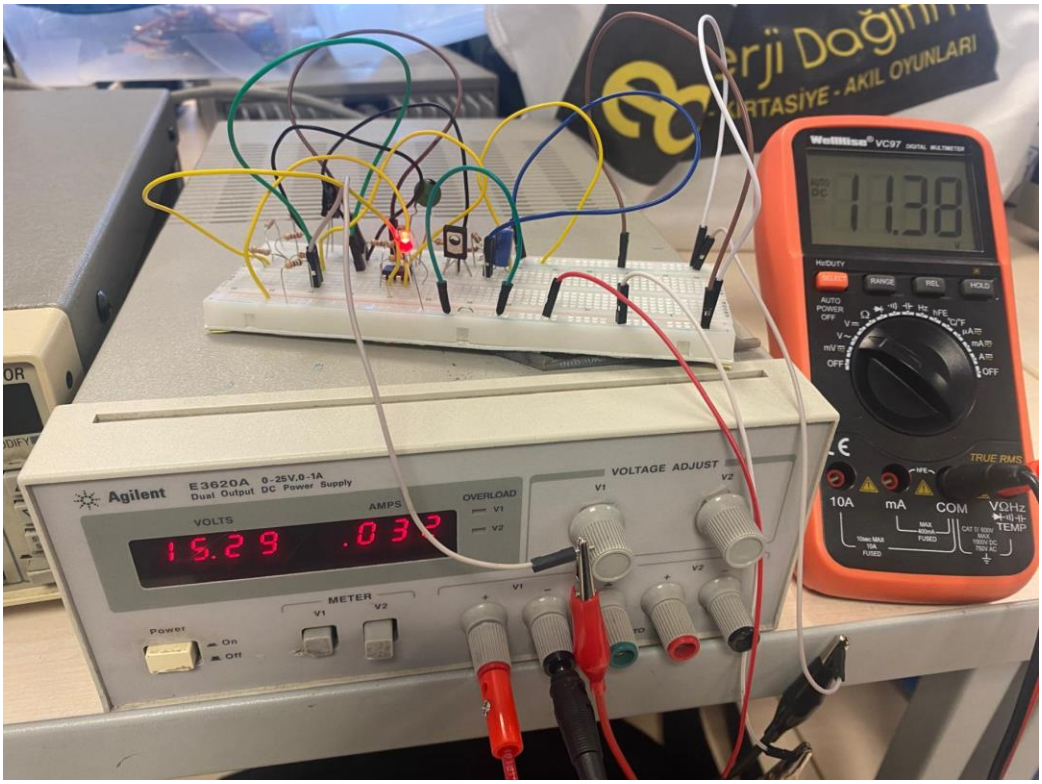


Figure 24: Regulation is achieved under 15.3V

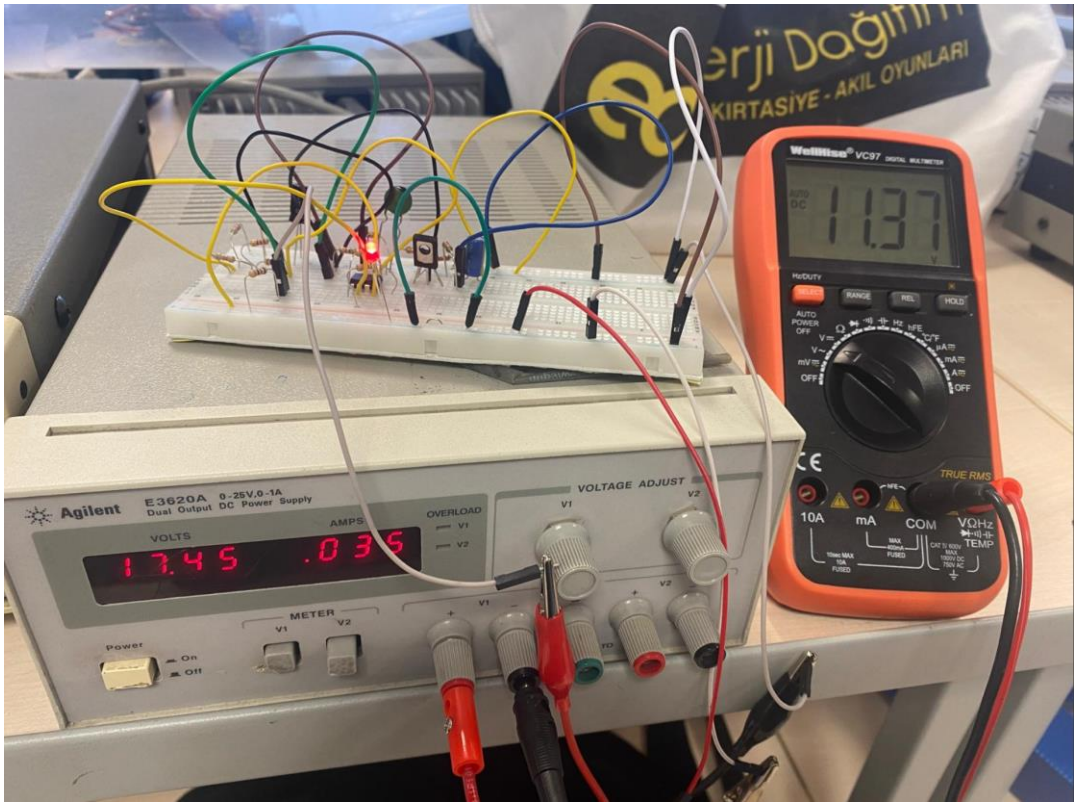


Figure 25: Regulation is achieved under 17.45V

We can see from these images that the output voltage is between 11.37V and 11.38V. The difference is smaller than 10mV so out first requirement is successful.

2) Load regulation. Changes of the load current should NOT change the value of the output voltage under the same input voltage. Out input voltage is 13.5V in this case. To change the load current, we basically remove the load resistor and expect that the output voltage would remain the same. Here is the images of the measurements:

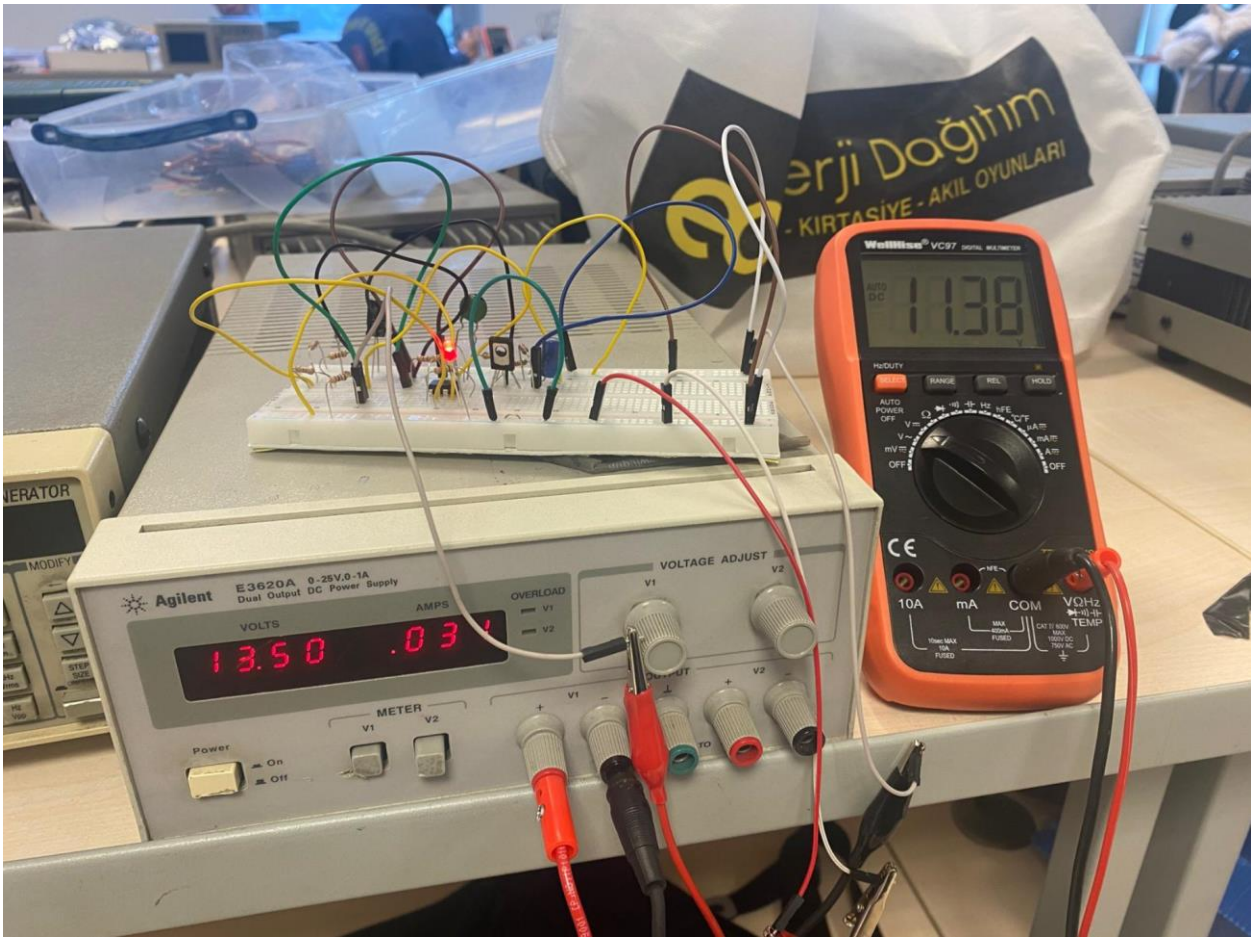


Figure 26: Output voltage with the load resistor.

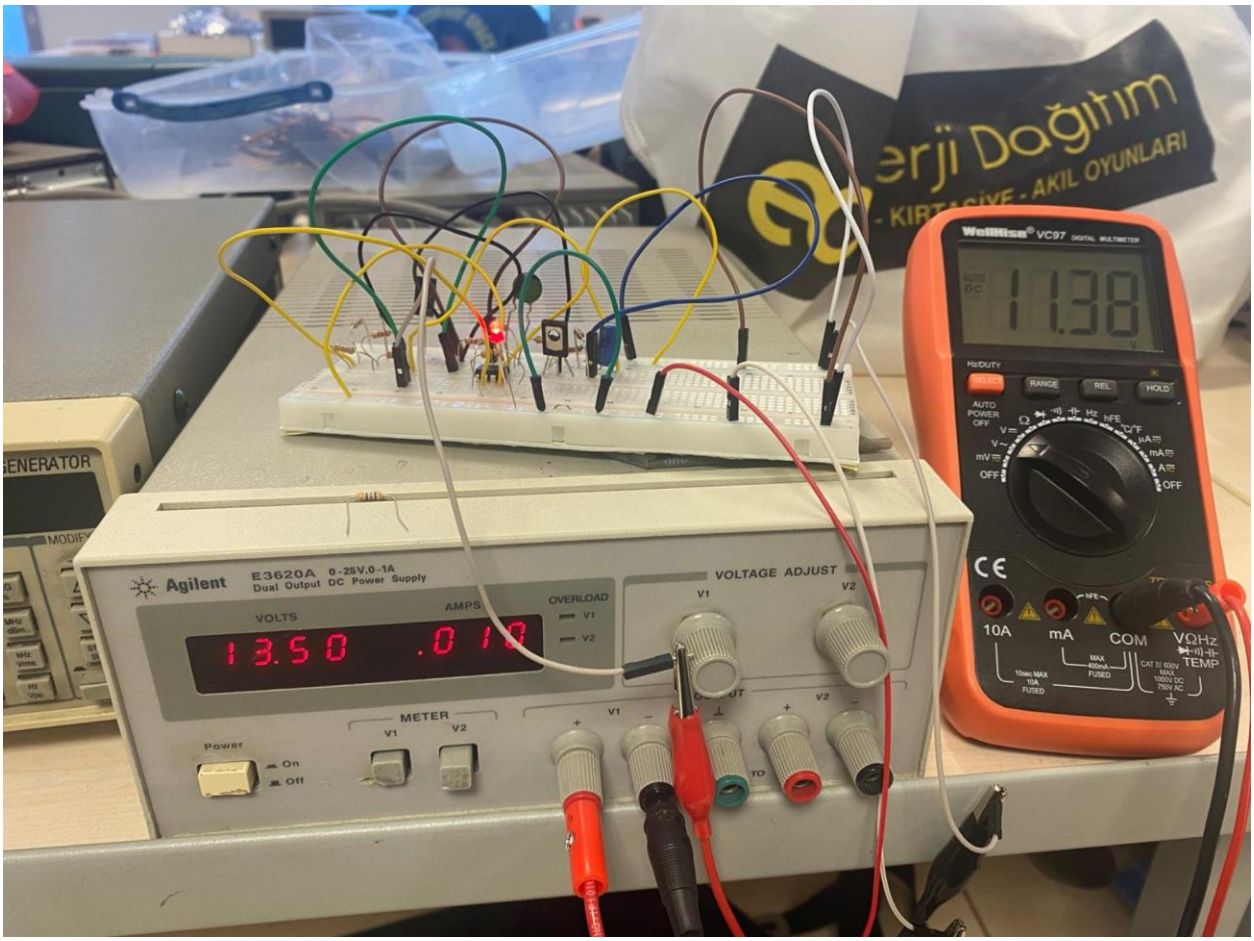


Figure 27: Output voltage with the load resistor removed.

We can see from these 2 images that the output voltage remains the same at 11.38V so we can say that the second requirement is also satisfied.

3) LED ON/OFF depending on regulation.

As it can be seen from Figure 21 to 27, we can see that while in regulation, the LED is ON. To satisfy the last requirement, we need to see some other situations where the circuit is NOT regulated. We can see if it is being regulated by looking at the output voltage. If it is reducing while the input voltage is being reduced, it is not regulated. If the LED is OFF in these situations, the requirement would be satisfied. Here are some images where the regulation is NOT achieved and the LED is OFF:

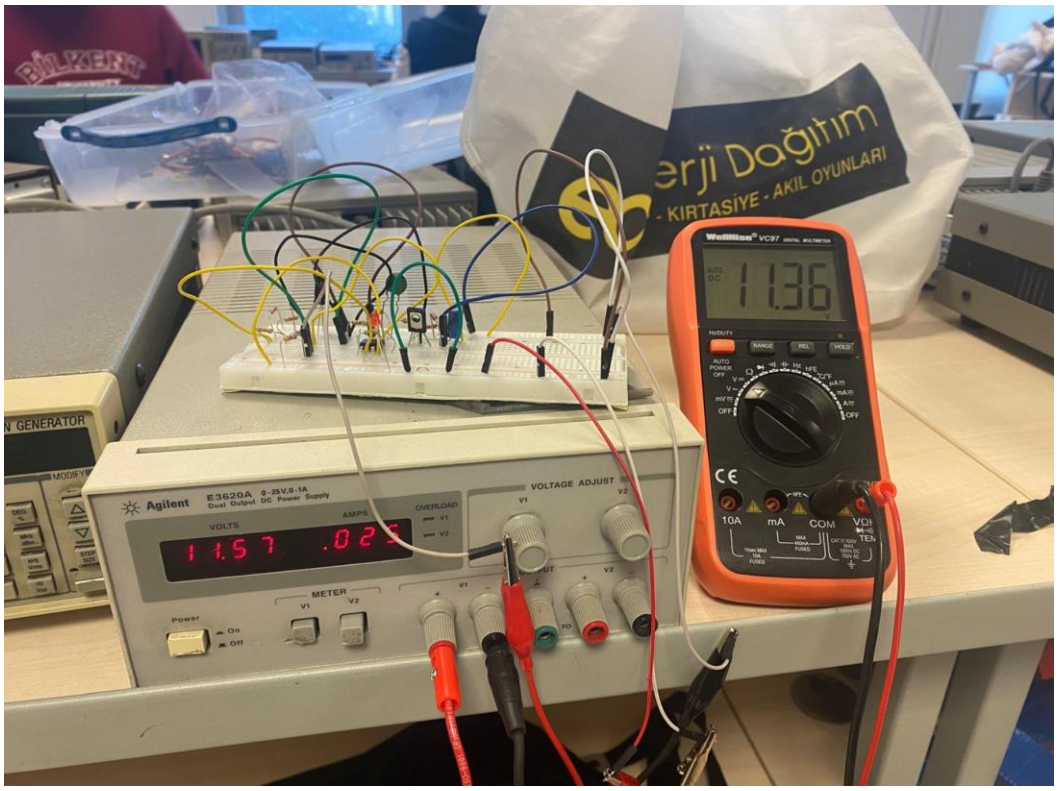


Figure 28: LED is OFF on input voltage of 11.57V

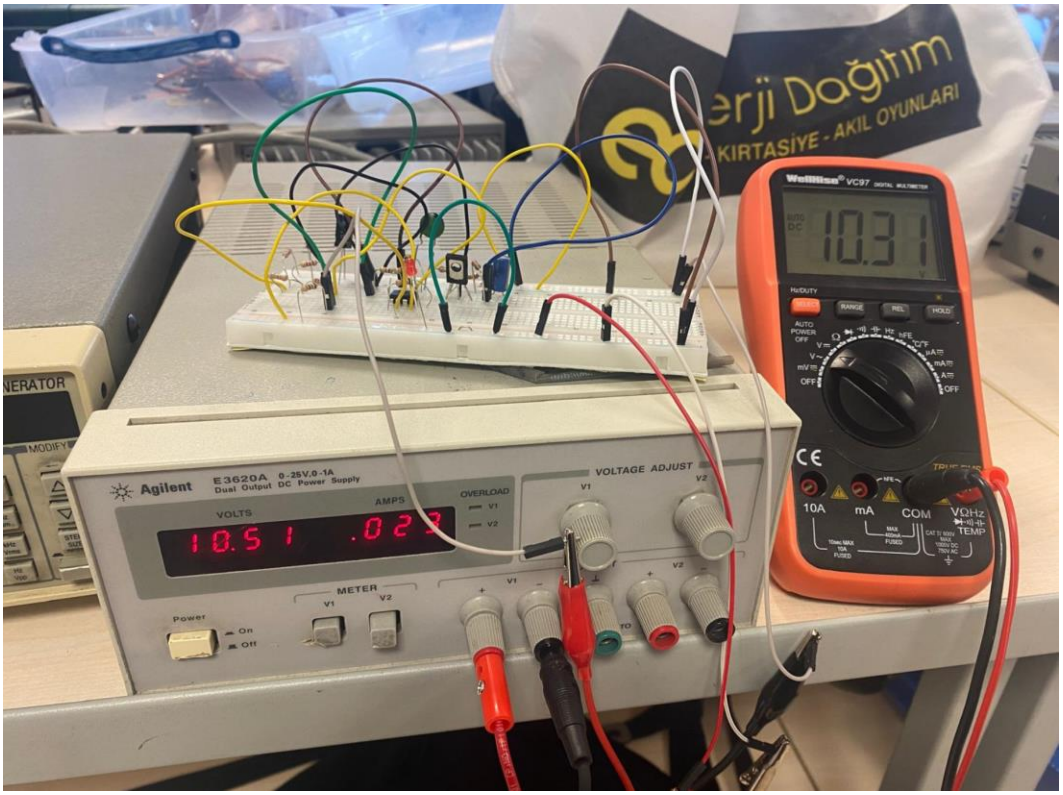


Figure 29: LED is OFF on input voltage of 10.5V

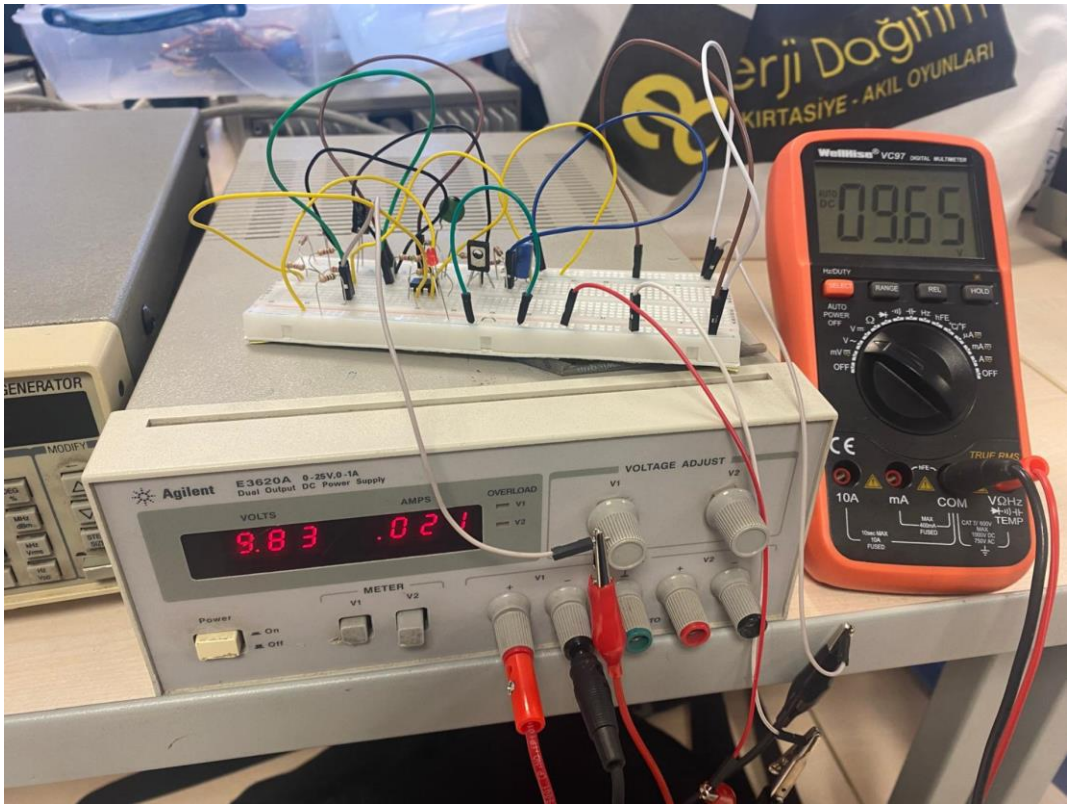


Figure 30: LED is OFF in the input voltage of 9.83V

Thus, the last requirement is also satisfied.

Conclusion

To conclude, on this lab experiment, the source and the load regulation were examined, by using the specifications of a Zener diode and a pnp transistor. The hardware part was a bit tricky since we expect a good regulation, so the component values should be really close to the ideal calculations. Since it is not the case in general, it was a bit difficult to see the required output voltage value. By making small adjustments, the required circuit is built and all of the three requirements were satisfied.